

Zachary Fried

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Education

University of Maryland

B.S. in Computer Science, second bachelor's degree; GPA: 3.97

- Coursework: Algorithms, Discrete Structures, Computer Systems, Advanced Data Structures, Database Design

College Park, MD

Expected May 2028

Oregon State University

B.S. in Chemistry

Corvallis, OR

Sept. 2015 - Sept. 2019

Experience

Textron Specialized Vehicles

June 2026 - August 2026

Software Engineering Intern

Augusta, GA

- Built backend features for a user-facing vehicle-tracking capability in a C#/.NET fleet-management platform, including an enablement toggle and configurable per-site vehicle limits persisted across site-level and global settings databases.
- Extended the same platform with a UDP request/response feature that decodes position requests from remote devices into C# objects, runs lookup logic to locate the relevant vehicles, and encodes and returns responses.
- Designed and built a reproducible evaluation harness for an internal application, comparing six self-hosted and hosted LLMs across 159 cases and grading tool calls against ground-truth state rather than model replies.
- Contributed to relational schema design with engineering mentors, proposing changes adopted into the final design.
- Built and browser-tested an internal business application from scratch using C#, SQL, JavaScript, Knockout.js, HTML/CSS.
- Implemented guided workflow logic mapping user selections to required outputs, replacing manual cross-referencing.

Axle Informatics / National Institutes of Health (NIH)

May 2024 - Aug. 2025

Linux Administrator and Data Processor (Contractor)

Baltimore, MD

- Root-caused recurring server crashes with custom log-capture utilities, later deployed as Nagios plugins, exposing a GPU-designed ML tool running CPU-only; fixed via CUDA configuration and per-user resource limits.
- Developed Bash, Python, and MySQL scripts to automate backups, file-integrity monitoring, and data processing.
- Deployed Docker containers for web applications and authentication services.
- Authored installation guides and system-configuration documentation adopted by multiple research labs.

National Institute on Drug Abuse (NIDA / NIH)

Aug. 2022 - May 2024

Postbaccalaureate Research Fellow

Baltimore, MD

- Engineered Python workflows to normalize, tokenize, and extract dictionaries from large-scale Reddit datasets.
- Optimized scripts transforming smartphone sensor data for a digital-biomarker analysis platform.

Projects

XNAT Deployment Toolkit | *Linux, Bash, Docker*

GitHub

- Built Bash and Docker tooling and recovery documentation for migrating a 7+ TB XNAT imaging platform.

Paper Meta-Analyzer | *Python, OpenAI API, PDF processing, MySQL*

GitHub

- Developed a Python pipeline that screens research papers via the OpenAI API and stores results in MySQL.

Publications

Co-author, "Evaluating generative AI responses to real-world drug-related questions," *Psychiatry Research*, 2024.

Co-author, "Exploring Sexualized Cannabis Use," presented at IC2S2, ICWSM, and CPDD, 2024.

Technical Skills

Languages: Python, C#, SQL, Bash, JavaScript, HTML/CSS; coursework: Java, C, Rust, OCaml

Infrastructure: Unix/Linux, Docker, Git, Azure DevOps, Visual Studio

Data: MySQL, SQLite, schema design, API integration, data pipelines